

DNA Printing Press

NUCLEUS
CYTOPLASM
ASYNCHRONY
DNA Printing Press
MEGALOBLASTS Nucleus
HYPERCELLULAR
SHREDDER
INEFFECTIVE
GAPS
DNA BREAKS

RATE-LIMITING
dUMP → dTMP

5,10-CH₂-THF MISSING
5,10-CH₂-THF

Methylfolate Trap

METHYL-B12
METHIONINE SYNTHASE
HOMOCYSTEINE METHIONINE
THF
POLYGLUTAMATE
SERUM FOLATE ↑ · CELL FOLATE ↓

↑ SERUM
5-MTHF
5-MTHF
TRAPPED
METHIONINE SYNTHASE
EMPTY
THF STARVATION
↓ CELL
SERUM FOLATE ↑ · CELL FOLATE ↓

METHYLATION ENGINE + CLASSIFICATION BOARD

>100 REACTIONS
METHYLATION CYCLE
METHIONINE
SAM Universal Methyl Donor
SAH
CH₃
B6
CYSTATHIONINE
CYSTEINE
↑ THROMBOSIS
B12 + FOLATE

CLASSIFICATION BOARD

HYPERSEGMENTED = MEGALOBLASTIC

MEGALOBLASTIC	NON-MEGALOBLASTIC
B12 FOLATE OROTIC ACIDURIA	ALCOHOL LIVER THYROID PREGNANCY MDS MYELOMA APLASTIC SMOKING
RARE CASE PANCREAS EAR w/ X RINGED SIDEROBLAST	
THIAMINE-RESPONSIVE · SLC19A2	

1. DNA printing press

WHAT YOU SEE

Printing press stamping DNA strands

WHAT IT ENCODES

Megaloblastic anemia = impaired DNA synthesis with normal RNA/protein (cytoplasmic) maturation, producing nuclear-cytoplasmic asynchrony: the cytoplasm 'matures' (hemoglobinizes) on schedule while the nucleus stays large, lacy and immature because it cannot finish replicating DNA. The defect is in the marrow precursors, so the cell line affected is all rapidly dividing tissue (marrow, GI epithelium). MCV is classically >100 fL and often >110-120 fL in florid B12/folate deficiency. Think 'big immature nucleus, mature cytoplasm' as the unifying lesion.

2. dUMP → dTMP rate-limiting

WHAT YOU SEE

Top box 'RATE-LIMITING dUMP→dTMP'

WHAT IT ENCODES

Thymidylate synthase converts dUMP→dTMP (the rate-limiting step that supplies thymidine for DNA), and it requires 5,10-methylene-THF as the one-carbon donor. When methylene-THF is unavailable, the cell substitutes dUTP for dTTP, triggering repeated futile excision-repair, DNA strand breaks and apoptosis of precursors. This is exactly the step targeted by 5-fluorouracil (inhibits thymidylate synthase) and indirectly by methotrexate (depletes reduced folate), so antimetabolite chemo reproduces megaloblastic morphology. Mnemonic: 'dUMP needs methylene-folate to become dTMP — no folate, no thymidine, no DNA.'

3. 5,10-CH2-THF missing

WHAT YOU SEE

'5,10-CH2-THF MISSING'

WHAT IT ENCODES

Depletion of 5,10-methylene-THF is the FINAL COMMON biochemical lesion of both B12 and folate deficiency — which is why both produce identical megaloblastic marrow and macrocytosis. In folate deficiency the donor is simply absent; in B12 deficiency it is functionally trapped as 5-methyl-THF (methylfolate trap). This convergence is the boards' favorite teaching point: morphology cannot distinguish B12 from folate — you need MMA/homocysteine and the neuro exam. Mnemonic: 'two roads (B12, folate), one dead end (no methylene-THF).'

4. Megaloblasts / asynchrony

WHAT YOU SEE

Newspaper print 'MEGALOBLASTS', 'ASYNCHRONY'

WHAT IT ENCODES

Marrow shows megaloblasts: abnormally large erythroid precursors with open, sieve-like ('lacy') chromatin and a nucleus that lags behind a hemoglobinized cytoplasm (nuclear-cytoplasmic asynchrony). Giant metamyelocytes and band forms are the granulocytic counterpart and are highly specific. Peripheral counterpart = oval macrocytes plus hypersegmented neutrophils (>5 lobes). Because all lines are affected, expect a mild pancytopenia, not just anemia. Mnemonic: 'megaloblast = young nucleus trapped in an old cell's body.'

5. Hypercellular ineffective marrow

WHAT YOU SEE

'HYPERCELLULAR', shredder 'INEFFECTIVE'

WHAT IT ENCODES

The marrow is HYPERcellular yet peripheral counts are low — the paradox of ineffective erythropoiesis: precursors are made in abundance but die in the marrow (intramedullary apoptosis) before release. This intramedullary hemolysis explains the lab triad of markedly elevated LDH, raised indirect bilirubin and low haptoglobin with a LOW reticulocyte count (the marrow is busy but unproductive). Reticulocyte index <2% with pancytopenia + hypercellular marrow points to a maturation defect (B12/folate, MDS) rather than blood loss.

BOARD TRAP

Hypocellular ('empty') marrow — that is aplastic anemia (pancytopenia with FATTY, hypocellular marrow and low retics). Megaloblastic marrow is the opposite: jam-packed/hypercellular. A 'pancytopenia + low retic' stem tempts you toward aplasia, but the hypercellular biopsy and giant metamyelocytes rule it in as megaloblastic.

6. Methylfolate trap

WHAT YOU SEE

Header 'Methylfolate Trap'

WHAT IT ENCODES

The methylfolate (methyl-THF) trap explains why B12 deficiency causes a FOLATE-type marrow lesion. Dietary folate enters cells and is reduced/methylated to 5-methyl-THF; the ONLY reaction that frees it back to usable THF is methionine synthase, which needs B12. Without B12 the entire folate pool 'piles up' as 5-methyl-THF and cannot be converted to the methylene-THF needed for thymidine synthesis. Net effect: functional intracellular folate deficiency despite normal/high body folate. Mnemonic: 'folate goes IN as methyl-THF and gets STUCK without B12 to let it out.'

7. Methionine synthase needs B12

WHAT YOU SEE

Methionine synthase machine, methyl-B12, homocysteine→methionine

WHAT IT ENCODES

Methionine synthase is one of only two B12-dependent human enzymes; it uses methylcobalamin to transfer a methyl group from 5-methyl-THF onto homocysteine, generating methionine AND regenerating free THF. The other B12 enzyme is methylmalonyl-CoA mutase (uses adenosylcobalamin; its block raises MMA — the neuro/myelin-relevant pathway). Knowing the two enzymes explains the labs: blocked methionine synthase → high homocysteine + folate trap; blocked mutase → high MMA, specific to B12. Mnemonic: 'B12 has two jobs — Methionine synthase (homocysteine) and Methylmalonyl mutase (MMA).'

8. 5-MTHF trapped, THF starvation

WHAT YOU SEE

'TRAPPED', 'EMPTY', 'THF STARVATION'

WHAT IT ENCODES

With methionine synthase stalled, 5-methyl-THF accumulates (the 'trapped' pool) while the metabolically active THF/methylene-THF forms run empty — the cell starves for usable folate even though total folate looks adequate. This is why giving FOLIC ACID alone to a B12-deficient patient transiently corrects the anemia (it floods the system enough to bypass the trap for DNA synthesis) but does NOTHING for the MMA/myelin pathway, allowing neuro damage to progress. Practical rule on boards: never replete folate without first excluding/treating B12.

9. Serum folate up, cell folate down

WHAT YOU SEE

'↑SERUM FOLATE ↓CELL FOLATE'

WHAT IT ENCODES

In B12 deficiency the trapped methyl-THF spills into serum, so SERUM folate can be normal or even HIGH while intracellular (RBC) folate is depleted — a classic discordance. RBC folate is the truer marker of tissue folate stores because it reflects the prior ~120 days, not the last meal. If a stem gives a high serum folate with a megaloblastic picture and neuro signs, suspect B12 deficiency, not folate excess. Mnemonic: 'serum lies, RBC folate tells the truth.'

10. Homocysteine ↑ thrombosis

WHAT YOU SEE

'↑HOMOCYSTEINE ↑THROMBOSIS', cystathionine→cysteine, B6

WHAT IT ENCODES

Both B12 and folate deficiency raise homocysteine (because methionine synthase needs both), making homocysteine SENSITIVE but NON-specific. Homocysteine is cleared by two routes: remethylation (B12/folate) and transsulfuration via cystathionine β-synthase to cysteine, which requires vitamin B6 (pyridoxine). Hyperhomocysteinemia is prothrombotic (endothelial injury, arterial+venous events) and is the link to homocystinuria. Boards pairing: high MMA + high homocysteine = B12; normal MMA + high homocysteine = folate.

11. Hypersegmented = megaloblastic

WHAT YOU SEE

'HYPERSEGMENTED ≠ MEGALOBLASTIC' classification arrow

WHAT IT ENCODES

Hypersegmented neutrophils (≥5 lobes, or ≥5% of neutrophils with 5 lobes / any with 6) are the most useful peripheral clue that a macrocytosis is MEGALOBLASTIC (B12, folate, antifolate drugs, hereditary orotic aciduria). They appear early and can persist after treatment. Their presence steers you away from the non-megaloblastic mimics. Mnemonic: 'big oval cells PLUS hypersegmented PMNs = think B12/folate first; order MMA + homocysteine.'

12. Non-megaloblastic macrocytosis

WHAT YOU SEE

List 'NON-MEGALOBLASTIC: ALCOHOL, LIVER, THYROID, MDS, MYELOMA, APLASTIC, SMOKING'

WHAT IT ENCODES

Macrocytic but NOT megaloblastic (no DNA-synthesis defect, NO hypersegmented neutrophils): alcohol (most common cause of mild macrocytosis), chronic liver disease (round target-rich macrocytes), hypothyroidism, MDS (often with cytopenias/dysplasia), reticulocytosis from acute hemolysis or blood loss (young large reticulocytes raise MCV), and drugs. A normal MMA/homocysteine plus normal smear lobe count points here. Mnemonic 'FAT RBC': Fetus/pregnancy, Alcohol/liver, Thyroid, Reticulocytosis, B-vitamin... but the discriminator is the SMEAR.

BOARD TRAP

Calling these 'megaloblastic.' They are NOT — there is no impaired DNA synthesis, so the marrow lacks giant metamyelocytes and the blood lacks hypersegmented PMNs. The single best discriminator is the smear (± MMA/homocysteine): hypersegmentation → megaloblastic (B12/folate); round macrocytes/targets without hypersegmentation → alcohol/liver; oval macrocytes with dysplasia/cytopenias → MDS.

13. Rare: thiamine-responsive

WHAT YOU SEE

'RARE CASE ... THIAMINE-RESPONSIVE SLC19A2'

WHAT IT ENCODES

Thiamine-responsive megaloblastic anemia (TRMA, Rogers syndrome) is a rare AR disorder from SLC19A2 mutations (defective high-affinity thiamine transporter) presenting with the triad of megaloblastic anemia + sensorineural deafness + diabetes mellitus, responsive to pharmacologic thiamine (B1). It is a board 'zebra' that explains a megaloblastic smear with NORMAL B12 and folate. Mnemonic: 'megaloblastic anemia that won't respond to B12/folate but DOES respond to thiamine — and the kid is deaf and diabetic.'